



BITS Pilani
DIGITAL

READER NOTES

FUNDAMENTALS OF FINANCE | MODULE 12

Data Science and Financial Modelling: Applications and Cases (Bond)



STUDY NOTES • EDITION 1.0 • WEEK 12

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How to use these notes



These notes cover the whole of Week 12: every concept and every worked example, written for reading and recall rather than for listening. They are yours to keep, mark up and revise from.

The week has nine topics in three parts. Part A confronts the problem that bond data mostly does not exist, and covers how to find and clean what does. Part B fills the gaps — pricing silent bonds, reducing the yield curve to three drivers, and understanding how a model chooses its numbers. Part C turns to credit risk. Read the parts in order; each assumes the last.

When assessment approaches you will not have time to re-read everything, and you will not need to. The Quick Recap boxes, the module summary and the self-check at the back were written for exactly that week. A workable plan: recaps first, then the summary, then attempt the self-check from memory before looking at the answers.

What each topic contains

Every topic follows the same rhythm, so you always know where to look.

Section	What it gives you
In one line	The whole idea compressed to a single sentence
Why this matters	Where the concept sits in the course, and why it earns your attention
The core idea	The full explanation, developed step by step
Figure	A diagram or comparison that makes the idea visible
Callouts	Key ideas, worked examples, pitfalls and exam pointers
Quick recap	Five bullets to revise from

Callouts used in this document

KEY IDEA

The most important takeaway of a section. If you remember nothing else, remember this.

COMMON PITFALL

A mistake learners make repeatedly, in assignments and in examinations.

EXAM POINTER

How the concept is typically tested, and what a full answer must contain.

REAL-WORLD EXAMPLE

A concrete case showing the concept at work outside the classroom.

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CONSOLIDATION

Module summary, by learning objective

Glossary and self-check

Module overview



Week 11 had too much information — hundreds of signals arriving faster than anyone could read them, so the machine helped us *choose*. This module reverses the problem. Bond prices are missing and trades are rare, so the machine helps us *see*: estimating a price where no trade occurred, and saying honestly how confident it is about each estimate.

One thread runs through all nine topics: **an estimate is only useful alongside its uncertainty**. Flag what was modelled rather than observed, publish a confidence band, and read a classifier by its four counts rather than its headline accuracy.

A note on coverage: the lecture material refers to interest rate forecasting as a further application of the methods in Topic 6. No recorded video covers it, so it is not treated here.

Learning objectives

By the end of this module, you will be able to:

- 1 **Explain** why bond data is scarce, choose the right data shape, and describe the six cleaning steps.
- 2 **Estimate** the yield of a bond that did not trade, and state its uncertainty honestly.
- 3 **Describe** the three movements of the yield curve and explain how maximum likelihood selects model settings.
- 4 **Explain** classification for default prediction and read a confusion matrix correctly.

Topic map

9 topics, three parts. The Source column gives the lecture video each topic draws on, so you can go back to the original; the Serves column shows which objective it supports.

Topic	What it covers	Part	Source	Serves
1	The silence problem	A • The data	L39 V1	LO 1
2	Three shapes of data, and where it comes from	A • The data	L39 V2	LO 1
3	Why data arrives dirty	A • The data	L39 V3	LO 1
4	Pricing bonds that did not trade	B • Filling gaps	L39 V4	LO 2
5	Three movements that explain a whole market	B • Filling gaps	L39 V5	LO 3
6	How a model chooses its numbers	B • Filling gaps	L39 V6	LO 3
7	A rating is a photograph	C • Credit risk	L39 V7	LO 4
8	Classification and logistic regression	C • Credit risk	L39 V8	LO 4
9	The confusion matrix and rating migration	C • Credit risk	L39 V8	LO 4

Key terms at a glance



Fuller definitions appear in the glossary at the back. Skim these before your first read; they carry most of the week's vocabulary.

Cross-sectional data

Many items observed at one point in time — a snapshot of the market.

Panel data

Many items across many dates; the richest and most demanding structure.

Web scraping

Programmatically reading a web page to extract data; must respect the site's terms of use.

Matrix pricing

Estimating a silent bond's yield from comparable bonds that did trade.

Yield curve

The yield the market demands at each maturity, from a few months to thirty years.

Nelson-Siegel

A yield curve formula using few parameters but flexible enough for real shapes.

Principal component analysis

An unsupervised technique reducing many related variables to a few drivers.

Maximum likelihood

Choosing the model settings that make the observed data most probable.

Logistic regression

A classification method using an S-shaped curve to keep probabilities between 0 and 1.

Threshold

The predicted probability above which a bond is treated as too risky.

False alarm

A safe bond classified as risky — costs forgone yield.

Rating migration matrix

The probabilities of moving from one rating to another over the next year.

Time series data

One item observed across many dates — the life story of one bond.

ISIN

International Securities Identification Number — the unique code assigned to a bond.

Covenant

A promise made by the borrower to the lender, whose breach may signal approaching distress.

Imputation

Estimating a missing value using patterns learned from available values.

Bootstrapping

Building the yield curve step by step from the shortest maturity outwards.

Confidence band

A range around an estimate showing how certain the model is.

Level, slope, curvature

The three movements explaining roughly 70%, 20% and 7% of yield curve variation.

Classification

Supervised learning where the predicted answer is a category rather than a number.

Interest coverage ratio

Profit divided by interest payable; a low value signals difficulty meeting obligations.

Confusion matrix

A table counting correct classifications, false alarms and missed defaults.

Missed default

A risky bond classified as safe — costs principal.

PART A • GETTING THE DATA

When the prices do not exist

1 The Silence Problem

PART A • GETTING THE DATA • TOPIC 1 OF 9 • SERVES LEARNING OBJECTIVE 1



IN ONE LINE

Two hundred and forty of Arvind's 268 bonds did not trade even once today. He must still publish a value for every one of them tonight.

Why this matters

Week 11 had too much information. This week has too little, and that reversal changes what the machine is for — from choosing among signals to carefully filling in what is missing.

Three hundred crore, and nobody rang a bell

You manage a bond fund worth ₹10,000 crore. You arrive at nine in the morning, buy nothing and sell nothing. By half past three the portfolio is worth ₹9,700 crore. Three hundred crore has disappeared. No company defaulted, no coupon was missed. Only the market interest rate changed.

Week 8 explained that much. Here is the part it did not. Arvind's portfolio holds **268 different bonds**. He opens the day's trading data and finds that only **28** of them traded. The remaining 240 did not trade even once — not a single buyer, not a single seller, all day.

So what is a bond worth when nobody bought it and nobody sold it? Not what it was worth last month — what it is worth *tonight*. A valuation must be published, because tomorrow morning thousands of investors will buy and sell units of the fund using it.

Why the missing prices matter

Week 7 established that a bond is a promise and that prices and rates move inversely. Week 8 followed Rahul's bond and developed duration, immunisation and the matching of a portfolio's duration to a future liability.

Every one of those ideas needs one input: **a current, trustworthy yield**. Arvind does not have it for most of his bonds tonight. The mathematics is elegant, but it depends entirely on market information that is often unavailable.

How the desk works today

At six in the evening two analysts start on the 240 silent bonds. They search spreadsheets for recent trades in comparable bonds and review the largest holdings carefully. For the smaller ones, time runs out — they carry yesterday's price forward and move on.

Nobody has time to work out which final numbers are reliable and which rest on weak assumptions, and the valuation is published as **one confident column of figures**. Meanwhile the credit team reviews borrowers in detail only once or twice a year, so many are being monitored on information several months old.

EQUITY ABUNDANCE, BOND SILENCE

THE EQUITY MARKET

INSTRUMENTS

50 stocks

TRADING FREQUENCY

Each may trade every few seconds

PRICES PRODUCED IN A DAY

Roughly 3,50,000 separate prices

THE INVESTOR'S SITUATION

Surrounded by numbers

THE BOND MARKET

INSTRUMENTS

268 listed bonds

TRADING FREQUENCY

Only 28 traded all day

PRICES PRODUCED IN A DAY

240 bonds produced none at all

THE INVESTOR'S SITUATION

Surrounded by silence

*The machine's job reverses. In Week 11 it helped Meera **choose** among too many signals. Here it helps Arvind **see** — estimating a price where no trade occurred, and stating how confident it is.*

Figure 1.1 The same trading day, two entirely different data problems.

KEY IDEA

A missing price is not a small inconvenience to be patched over. It is the **normal condition** of the bond market, and any valuation process that does not confront it directly is publishing guesses dressed as observations.

EXAM POINTER

State the contrast numerically — 28 traded against 240 silent — and explain why it matters: every technique from Weeks 7 and 8 requires a current yield. The examinable insight is that the elegance of the mathematics depends on market data that frequently does not exist.

QUICK RECAP • TOPIC 1

- A bond portfolio can lose ₹300 crore in a day with no default and no trade.
- Of 268 bonds, 28 traded and 240 produced no price at all.
- Equities produce hundreds of thousands of prices a day; bonds produce almost none.
- Every technique from Weeks 7 and 8 needs a current, trustworthy yield.
- Today's desk carries stale prices forward and publishes them without flagging their reliability.

2 Three Shapes of Data, and Where It Comes From

PART A • GETTING THE DATA • TOPIC 2 OF 9 • SERVES LEARNING OBJECTIVE 1



IN ONE LINE

Choosing the wrong shape is among the most common beginner mistakes — and finding the data takes four-fifths of a real project.

Why this matters

Week 11 assumed the data was available. This week, obtaining it is a major part of the challenge, and the shape you need follows from the question you are asking.

Decide the shape first

Ask any professional on a real data project and you will hear the same thing: finding, cleaning and preparing data takes roughly **four-fifths** of the total time, and the modelling takes the remaining fifth.

Before collecting a single number, decide what shape your question requires. Picture each as a grid where every row is a bond and every column a date. **Cross-sectional** data takes one column — many items at one moment. **Time series** takes one row — one item across many dates. **Panel** data takes the whole grid, which is why it is the most demanding to collect and manage.

Where the data comes from

Exchange and depository files: the NSE and BSE publish daily information on traded bonds, and clearing organisations publish settlement prices, often as CSV files. **Regulator and central bank websites:** the RBI publishes policy rates, auction results and yield curve information. **Credit rating agency publications:** Crisil, ICRA and CARE publish rating changes and default statistics — and these default records become the known outcomes used to train a prediction model. **Commercial data vendors:** paid services that collect, clean and organise data already prepared for analysis.

How it reaches your computer

From simplest to most sophisticated. A **manual CSV download** is acceptable when data is needed once, but inefficient daily. An **application programming interface** is a service counter through which a programme requests data and receives it automatically. **Web scraping** means writing a programme that reads a web page as a human would and extracts the table.

Scraping must be used responsibly — respect the site's terms of use, since thousands of automated requests place an unfair burden on a public service and are often specifically prohibited.

THREE SHAPES • THE QUESTION DECIDES WHICH ONE YOU NEED

Shape	Structure	Example	Used for
Cross-sectional	Many items, one point in time — a photograph of the market	Valuing all 268 bonds tonight	Arvind's portfolio valuation
Time series	One item, many dates — the life story of one bond	One bond's yield every week for five years	Understanding trends; predicting where rates may head
Panel	Many items across many dates — the richest and most demanding	200 bonds observed monthly for ten years	Which companies defaulted, and what warning signs came first

*Finding, cleaning and preparing data takes about **four-fifths** of a real project. The modelling takes the remaining fifth.*

Figure 2.1 The three data shapes. Panel data is the richest and by far the hardest to assemble.

COMMON PITFALL

Collecting the data first and deciding the question afterwards. If you need to know which warning signs preceded defaults, only panel data will do — and discovering that after assembling a cross-section means starting again.

EXAM POINTER

Name all three shapes with an example of each, and match a stated question to the right one. Name the four data sources and the three collection methods, and note the legal and ethical constraint on scraping — that point is frequently examined and often omitted.

QUICK RECAP • TOPIC 2

- Data preparation takes roughly four-fifths of a real project; modelling takes a fifth.
- Cross-sectional data is many items at one moment; time series is one item across many dates.
- Panel data is the whole grid — richest, most demanding, and needed for default studies.
- Sources: exchanges and depositories, regulators, rating agencies, commercial vendors.
- Collection runs from manual CSV download to APIs to web scraping, which must respect terms of use.

3 Why Data Arrives Dirty

PART A • GETTING THE DATA • TOPIC 3 OF 9 • SERVES LEARNING OBJECTIVE 2



IN ONE LINE

Cleaning is not clerical work that precedes the modelling. Every cleaning decision silently teaches the model what the world looks like.

Why this matters

Bond data is among the dirtiest in finance, and this is the topic where careless work does the most damage — invisibly, because the model will run perfectly well on corrupted inputs.

What a raw file looks like

A raw download may show the same bond twice, its name in capitals on one row and lowercase on another, so the computer fails to see they are the same. A price may be missing and appear as “not available”. Dates may appear in three different formats. There may be a blank row at the bottom — harmless to a person, but enough to interrupt an automated programme.

After cleaning, the same dataset should hold one row per bond, one consistent format in every column, and any estimated value **clearly marked** so nobody mistakes it for a real market price.

The step where beginners do the most damage

Step five — handling missing values — deserves special attention. **A missing price is not zero.** Replace blanks with zeros and the reported portfolio value may collapse on paper, while the model learns completely unrealistic relationships.

Three honest ways to handle a missing value: *remove* the row and disclose that it was removed; *carry* the last observed value forward and label it as such; or *estimate* it with a model and clearly flag it as an estimate. The third is the method examined in Topic 4.

The information that is not in the table

Much of what matters in bond markets sits in documents rather than tables — rating agency reasoning, offer documents, and the legal **covenants** attached to a bond.

A covenant is a promise made by the borrower to the lender: keep debt below three times profits, do not sell the main factory, restrict dividends above a set level. Breaking one may immediately give the lender additional rights — and may signal approaching distress. These documents run to hundreds of pages, and historically many were never reviewed again after purchase.

Language models can now read them, extract every covenant into a table, and compare each against the borrower’s latest financials every quarter — warning an analyst that a company may be three months from breaking a promise made five years earlier.

Cleaning is part of the modelling

Every cleaning decision teaches the model something. The rows removed matter. The treatment of missing prices matters. Whether a 45% yield is classified as an error or a genuinely distressed bond matters.

Two analysts can receive the same raw file and use the same algorithm and still produce different models, with the difference coming almost entirely from how each cleaned the data. Professional teams therefore document their cleaning rules and subject those rules to review.

One danger in particular: deleting every unusual row. The unusual rows may be genuinely distressed bonds — exactly the examples a credit risk model most needs to learn from. Deleting inconvenient observations teaches the model that financial trouble never happens.

SIX CLEANING STEPS • AND WHAT EACH ONE CATCHES

Step	What it fixes	Caught in the exercise below
1 • Fix the identifiers	Every bond has an ISIN, written identically throughout	The repeated ISIN; the 101.2 against 1012
2 • Fix the units	Rupees or paise; per cent or basis points — 7.9% and 790bp are the same value	The 101.2 against 1012
3 • Standardise dates	Convert every date into one format	—
4 • Remove duplicates	Identify and delete repeated records	The repeated ISIN
5 • Handle missing values	Never replace a blank by quietly typing a guess	The twelve zero prices
6 • Reasonableness check	Apply financial knowledge — a 45% corporate yield is probably an error	The three yields above 60%

The exercise. You download 500 corporate bonds. Twelve show a price of zero. Three show yields above 60%. The same ISIN appears twice. One row shows 101.2 and another 1012. Which step catches each?

Figure 3.1 The six cleaning steps, with the exercise mapped against them.

KEY IDEA

Data cleaning is **not preliminary to the modelling — it is part of it**. A model is shaped as much by which rows were discarded as by which algorithm was chosen, and only one of those decisions usually gets documented.

EXAM POINTER

Name all six steps in order and match problems to steps. State the three honest ways to handle a missing value. The highest-value point is that cleaning decisions shape the model — same file, same algorithm, different cleaning, different result.

QUICK RECAP • TOPIC 3

- Bond data arrives with duplicates, inconsistent names, missing prices and mixed date formats.
- Six steps: identifiers, units, dates, duplicates, missing values, reasonableness.
- A missing price is not zero; remove, carry forward, or estimate and flag.
- Covenants sit in documents rather than tables, and language models can now extract them.
- Two analysts with the same file and algorithm can produce different models through cleaning alone.

PART B • FILLING THE GAPS

Estimating what was never observed

4 Pricing Bonds That Did Not Trade

PART B • FILLING THE GAPS • TOPIC 4 OF 9 • SERVES LEARNING OBJECTIVE 2



IN ONE LINE

A small number of actual trades can supply estimated yields for many bonds that never traded — provided every estimate is flagged as one.

Why this matters

This solves the problem Topic 1 posed. It is also the clearest example in the course of a machine formalising something an experienced human already does by judgement.

How a dealer does it

Asked to value a bond that did not trade, an experienced dealer looks for comparable bonds — similar maturity, similar credit rating, similar industry — examines the yields at which those traded, and estimates from them.

A five-year bond issued by a AA-rated power company did not trade today. Three comparable five-year AA bonds traded at yields between 7.85% and 7.95%. The dealer may reasonably estimate **about 7.90%**. The market calls this **matrix pricing**.

A machine learning specialist describes the same process differently: features go in — maturity, credit rating, industry, issue size, coupon rate — and the model produces an estimated yield. Estimating a missing value using patterns learned from available values is called **imputation**.

Fitting the yield curve

Bonds have a structure that makes this far more powerful than estimating one at a time. Bonds of similar credit quality can be organised along a **yield curve**, showing the yield the market demands at each maturity from a few months to thirty years.

Picture a graph with maturity across the bottom and yield up the side. The eight bonds that traded today appear as real observed points. The model fits a smooth curve through them. Each silent bond's yield can then be read off the curve at that bond's own maturity — so a fifteen-year bond untraded all week still receives a fair and defensible yield.

In machine learning language this is another regression, built on the same principle as the straight line in Week 11 — except the relationship is a curve, because real yield curves bend. Two methods are named: **bootstrapping**, which builds the curve step by step from the shortest maturity outwards, and **Nelson-Siegel**, a formula using few parameters yet flexible enough for real shapes.

Two non-negotiable practices

Flag every estimate. A price produced by a model is not a price observed in a trade, and any report hiding that difference is misleading. Professional systems preserve the distinction throughout.

Publish a confidence band — a range around the estimate showing how certain the model is.

TWO ESTIMATES, TWO VERY DIFFERENT LEVELS OF CONFIDENCE

Bond	Situation	Estimated yield band	What it tells the committee
5-year, highest rating	Surrounded by comparable bonds that traded recently	7.88% to 7.92%	Strong evidence — the estimate can be relied on
15-year, small issuer	Bonds of this type rarely trade	8.4% to 9.6%	Treat with caution — the estimate rests on thin evidence

*The wide range is **not a weakness to be hidden**. It may be the most useful information in the report, telling the risk committee exactly which valuations rest on strong evidence and which do not.*

Figure 4.1 Confidence bands. A single number would have concealed the difference entirely.

KEY IDEA

The valuable output is not the estimate. It is the estimate *plus its uncertainty* — because a committee that knows which numbers are shaky can act on that, and a committee handed one confident column cannot.

EXAM POINTER

Explain matrix pricing in the dealer's language and imputation in the machine's, and state that they are the same process. Describe curve fitting, name bootstrapping and Nelson-Siegel with what each does, and give both professional practices — flag estimates, publish bands.

QUICK RECAP • TOPIC 4

- Matrix pricing estimates a silent bond's yield from comparable bonds that did trade.
- The machine learning name for the same process is imputation.
- Fitting a yield curve through observed trades supplies yields for every silent bond at its own maturity.
- Bootstrapping builds the curve outwards; Nelson-Siegel fits it with few parameters.
- Always flag model estimates as estimates, and always publish a confidence band.

5 Three Movements That Explain a Whole Market

PART B • FILLING THE GAPS • TOPIC 5 OF 9 • SERVES LEARNING OBJECTIVE 3



IN ONE LINE

Fifteen moving yields reduce to three underlying drivers — level, slope and curvature — which together explain about 97% of what the curve does.

Why this matters

Principal component analysis is the unsupervised technique that makes bond risk reportable. It also has a caution attached that the mathematics itself cannot supply.

The problem

Arvind does not monitor one interest rate but the whole yield curve — perhaps 15 maturities from three months to thirty years, every one moving daily. If he wants to know what happens to his portfolio when rates change, which of the 15 should he change, and by how much? They do not move independently.

Collect several years of daily changes and ask a machine to find the hidden structure, without giving it any predetermined categories — **unsupervised learning**. The result is remarkable: almost every movement of the yield curve can be represented as a combination of three basic movements.

Why a bond manager should care

Risk becomes reportable. Without PCA, Arvind may need to discuss 15 different interest rate sensitivities. With it, the risk report can say the portfolio loses ₹40 crore if the overall level of rates rises one percentage point, gains ₹12 crore if the curve steepens, and is only slightly affected by curvature. Three clear statements a committee can discuss.

Stress testing becomes realistic. A simple stress test raises every yield by one point, but markets rarely move equally at every maturity. PCA lets the manager stress the curve in ways consistent with how it has actually behaved.

Hedging becomes accurate. Duration mainly protects against a *level* movement. A portfolio holding long positions at one end and short at the other may look properly hedged on duration, yet still lose substantially if the curve steepens. PCA reveals those exposures.

The caution the mathematics cannot supply

PCA identifies the movements explaining the most variation, but **it does not name them**. The labels level, slope and curvature are assigned afterwards by finance professionals because the patterns match familiar behaviour. In another dataset the components may have no such intuitive meaning.

The machine does the numerical analysis; the professional must still interpret the results and decide whether to trust them.

THREE MOVEMENTS • ABOUT 97% OF EVERYTHING THE CURVE DOES

Movement	What happens to the curve	Share of historical movement
Level	The entire curve shifts up or down together. All maturities move the same way, so the shape stays similar and only the height changes	About 70%
Slope	Short and long yields move differently — in opposite directions, or one end far more than the other. The curve becomes steeper or flatter	About 20%
Curvature	The middle of the curve rises or falls relative to the two ends, bulging or sagging while the ends stay broadly unchanged	About 7%

***Exercise.** The RBI cuts the repo rate. Short-term yields fall half a percentage point; 30-year yields are almost unchanged. Which movement? **Slope** — the short end fell while the long end held, so the curve steepened. And note that a duration-hedged portfolio can still be affected.*

Figure 5.1 The three principal components of the yield curve, and how much each explains.

KEY IDEA

PCA is related to clustering. Clustering groups assets that behave similarly; PCA reduces many related *variables* to a few common drivers. Both are unsupervised, and both find structure nobody specified in advance.

EXAM POINTER

Name all three movements with their approximate shares — 70, 20 and 7 per cent, summing to about 97 — and describe what each does to the curve. Give at least two of the three practical benefits. The caution about naming is a distinguishing point: PCA finds components, humans label them.

QUICK RECAP • TOPIC 5

- Fifteen moving yields reduce to three underlying movements found by unsupervised learning.
- Level explains about 70%, slope about 20% and curvature about 7% — roughly 97% together.
- Risk reports become three statements a committee can actually discuss.
- Duration protects mainly against level movements; PCA reveals slope and curvature exposure.
- PCA finds the components but does not name them — the interpretation is human.

6 How a Model Chooses Its Numbers

PART B • FILLING THE GAPS • TOPIC 6 OF 9 • SERVES LEARNING OBJECTIVE 3



IN ONE LINE

Choose the settings that make the data you actually observed as probable as possible. That is maximum likelihood.

Why this matters

Week 11 gave one answer to how a model picks its numbers — least squares. This is the more general method, and it is what produces the confidence bands from Topic 4.

The idea in one sentence

Instead of asking how far the model lies from the data, ask: *if this model were true, how likely would today's data look this way?* Then choose the model under which the observed data is **least surprising**.

The coin, worked without algebra

You flip a coin ten times and observe seven heads. What should you estimate as the probability of heads?

If the true probability were 0.2, seven heads would be extremely unusual. If 0.5, possible but still somewhat fortunate. If 0.7, seven out of ten is exactly what we would expect on average. If 0.9, seven heads would be *lower* than expected — we would have anticipated about nine.

The value making the observed result most probable is **0.7**. That is the maximum likelihood estimate, reached without any advanced mathematics — simply by considering several possible values and picking the one under which the result is least surprising.

Applied to bonds, and why it produces the bands

A yield curve model contains a few settings controlling level, slope and curvature. Maximum likelihood asks which combination makes today's observed trades most probable. If eight bonds traded, it selects the settings under which those eight yields were most likely to have occurred — and that gives the fitted curve.

It also produces the **confidence bands**. A likelihood does not only identify the best-fitting answer; it shows how much better that answer is than nearby alternatives. If several nearby parameter values explain the data almost equally well, the estimate is uncertain and the band should be wide. If the best value explains the data much better than its neighbours, the band can be narrow. **The sharpness of the likelihood tells us how much confidence to place in the estimate.**

Why it handles messy reality better

Least squares usually assumes errors are symmetrical and well behaved. Maximum likelihood lets us state explicitly what kind of environment we believe produced the data — that defaults are rare events, that yield errors should be larger for bonds that rarely trade than for those trading frequently, that extreme movements occur more often than a simple bell curve suggests.

This matters particularly in bond markets, where data is limited, uneven and concentrated in a few actively traded securities. A yield from an actively traded bond is more dependable than an estimate from one that rarely trades, and maximum likelihood provides the framework for saying so.

FOUR CANDIDATE VALUES, ONE CLEAR WINNER

If the true probability were...	Then seven heads in ten would be...	Likelihood
0.2	Extremely unusual	About 0.0008
0.5	Possible, but somewhat fortunate	About 0.117
0.7	Exactly what we would expect on average	About 0.267
0.9	Lower than expected — we would have anticipated about nine	About 0.057

*The likelihood at 0.7 is more than **twice** that at 0.5 and nearly five times that at 0.9. How sharply it peaks is what determines whether the confidence band is narrow or wide.*

Figure 6.1 Maximum likelihood by inspection. The peak identifies the estimate; its sharpness sets the band.

KEY IDEA

Least squares asks how far the model is from the data. Maximum likelihood asks how *probable* the data is under the model — which is a richer question, because the answer carries its own measure of uncertainty.

EXAM POINTER

State the principle in one sentence and work the coin example by inspection across several candidate values. Explain how the sharpness of the likelihood produces the confidence band, and give one reason it suits bond data better than least squares — uneven reliability across securities.

QUICK RECAP • TOPIC 6

- Maximum likelihood chooses the settings making the observed data most probable.
- Seven heads in ten flips gives a maximum likelihood estimate of 0.7, found without algebra.
- Applied to a yield curve, it selects the level, slope and curvature settings that best explain observed trades.
- The sharpness of the likelihood produces the confidence band around the estimate.
- It suits bond data because it can state that some observations are more reliable than others.

PART C • CREDIT RISK

Who will pay, and who will not

7 A Rating Is a Photograph

PART C • CREDIT RISK • TOPIC 7 OF 9 • SERVES LEARNING OBJECTIVE 4



IN ONE LINE

Ratings change in steps; companies weaken continuously. The gap between the two is where losses occur.

Why this matters

Part C turns from valuation to credit. This topic frames why a model can add anything at all to work already done carefully by professional analysts.

What a rating actually is

When a rating agency assigns a company a AA rating, a team of analysts studies the business, meets management and forms a professional judgement. That is serious and valuable work — but it was completed on one particular day, possibly several months ago.

The photograph does not update when the company's largest customer cancels a contract in March. It does not update when the promoter quietly pledges shares in June. It does not update when the company's own bonds begin trading at yields strongly suggesting distress while the AA rating stands unchanged.

The gap

Ratings are revised occasionally, after committees meet and complete reviews. A company's financial condition can change every day. That gap between occasional updates and continuous change is where losses occur — and where machines can genuinely help.

What this is not

This is not a criticism of rating analysts. The point is narrower and more useful: **a professional judgement made periodically can be supplemented by a measurement updated daily.**

The rating remains the carefully prepared photograph. The machine provides the live video feed. Neither replaces the other, and a desk that discards the rating in favour of a model has misunderstood the argument as badly as one that ignores the model.

A PHOTOGRAPH AND A VIDEO FEED

THE CREDIT RATING

PRODUCED BY

A team of analysts studying the business and meeting management

UPDATED

Occasionally, after a committee review

STRENGTH

Deep, considered, contextual judgement

WEAKNESS

Fixed on the day it was made

THE MODEL

PRODUCED BY

Financial ratios, market signals and text, updated continuously

UPDATED

Daily, or faster

STRENGTH

Never stale; monitors thousands of borrowers at once

WEAKNESS

No contextual judgement of its own

Ratings change in **steps**; companies weaken **continuously**. A human analyst may monitor 15 borrowers at this level of detail. A machine can monitor 2,000.

Figure 7.1 Not a replacement. A periodic judgement supplemented by a continuous measurement.

KEY IDEA

The problem is not that ratings are wrong. It is that they are **discrete and infrequent** while the thing they describe is continuous — and no amount of analytical care fixes a mismatch of frequency.

EXAM POINTER

State the photograph analogy and the specific gap: occasional revision against continuous change. Be careful to frame the model as *supplementing* rather than replacing the rating — answers that dismiss rating agencies miss the argument being made.

QUICK RECAP • TOPIC 7

- A credit rating is a careful professional judgement made on one particular day.
- It does not update when circumstances change between reviews.
- Ratings change in steps; company condition changes continuously.
- That frequency mismatch is where losses occur.
- The model supplements the rating with a daily measurement; it does not replace the judgement.

8 Classification and Logistic Regression

PART C • CREDIT RISK • TOPIC 8 OF 9 • SERVES LEARNING OBJECTIVE 4



IN ONE LINE

Predicting a category rather than a number — and the output must be a probability, never a verdict.

Why this matters

Every question in Week 11 had a numerical answer. This one does not, which changes both the method and how the result must be reported.

A different kind of question

Arvind asks: will this borrower default during the next twelve months? The answer is yes or no. Predicting a category is called **classification**.

It is still *supervised* learning — the model learns from past examples where the answer is known. The difference is that the answer key holds categories rather than numbers, and the default records published by rating agencies supply those known outcomes.

Why a straight line will not do

The **interest coverage ratio** divides profit by interest payable. A low ratio means the company can barely afford its obligations. Now consider the probability of default within a year, which must always sit between 0 and 1.

Fit an ordinary straight line and it does not know where to stop. Extend it far enough one way and it produces a probability above 1; extend it the other way and it produces one below 0. Both are impossible.

Logistic regression uses an S-shaped curve to keep predicted probabilities between 0 and 1. Low coverage ratios push the probability towards 1, high ratios towards 0, and the curve changes most sharply in the uncertain middle range.

The output is a probability, not a verdict

This is the most important sentence in the topic. The model should say a borrower has an **8% probability** of default over the next twelve months. It should *not* say the borrower will default.

A borrower with an 8% probability is not automatically a bad borrower. It is one whose bond should be priced, and whose position should be sized, in a way that reflects 8% default risk.

But eventually someone must decide. That requires choosing a **threshold** — above what predicted probability is a bond too risky? Set it too low and the fund avoids many bonds that would have performed well; too high and it keeps bonds that eventually default. Choosing the threshold is a subjective but important professional decision.

Why explanations matter

An unexplained number is not sufficient for the borrower, the regulator or the investment committee. Credit decisions must be supported by understandable reasons, and supervisors increasingly require them to be explainable.

This is one reason **logistic regression remains widely used** despite more complex models existing: the weights it learns can be examined and interpreted. The model should not merely report a 12% probability — it should explain that the probability is driven mainly by the interest coverage ratio falling below 1.5 and by debt increasing for four consecutive quarters.

Explanations also catch model errors. If the model says a borrower is risky mainly because its ISIN begins with a particular sequence of characters, that reveals it has learned a meaningless pattern.

WHAT GOES INTO A DEFAULT MODEL

Category	Inputs	What they capture
Financial ratios	Interest coverage; debt-to-equity; cash against debt due within a year; margin trend	The borrower's capacity to service obligations
Market signals	The yield spread on its bonds; share price behaviour; share price volatility	What other investors already believe
Text signals	Earnings call language; the tone of recent news	What management and the press are signalling

*The same information a good credit analyst would examine. The difference is that **a human analyst may monitor 15 borrowers at this depth; a machine can monitor 2,000.***

Figure 8.1 The three input families. All three were available before — only the coverage is new.

COMMON PITFALL

Reporting the model's output as a decision. "This borrower will default" is not something the model can say. It produces a probability; converting that into an action requires a threshold, and the threshold is a human judgement about the relative cost of two different mistakes.

EXAM POINTER

Define classification as supervised learning with categorical answers. Explain why a straight line fails — it produces impossible probabilities — and that logistic regression uses an S-shaped curve. State that the output is a probability requiring a threshold, and give explainability as a reason logistic regression survives.

QUICK RECAP • TOPIC 8

- Classification predicts a category; it is supervised learning with a categorical answer key.
- A straight line produces impossible probabilities above 1 and below 0.
- Logistic regression uses an S-shaped curve to keep probabilities between 0 and 1.
- The output is a probability, not a verdict; converting it to action requires a threshold.
- Logistic regression persists because its weights can be explained to a committee and a regulator.

9 The Confusion Matrix and Rating Migration

PART C • CREDIT RISK • TOPIC 9 OF 9 • SERVES LEARNING OBJECTIVE 4



IN ONE LINE

A model that is 95.3% accurate still missed five of the eighteen defaults that occurred — which is why overall accuracy is the wrong measure.

Why this matters

The closing topic supplies the honest scorecard, and it repeats the Bayesian lesson of Week 11 in a credit setting: when the event is rare, accuracy misleads.

The scorecard

Despite its name, the **confusion matrix** is one of the clearest report cards available. It counts four outcomes, and the two diagonals mean very different things.

Reading it honestly

940 correct clearances and 13 correct warnings make 953 correct classifications out of 1,000 — an overall accuracy of **95.3%**. That sounds impressive.

But 18 bonds actually defaulted, and the model **missed 5 of them**. Missed defaults are especially dangerous, because they can create large losses. The 42 false alarms are safe bonds classified as risky, which may cause the investor to pass up good opportunities.

Here is the trap. Defaults are rare. If only 2% of bonds default in a year, a lazy model predicting that *no bond ever defaults* would be correct **98%** of the time — and completely useless, because it would identify not a single actual default. This is the same lesson Bayes taught in Week 11: when an event is rare, overall accuracy misleads.

And the two errors do not cost the same. A false alarm may cost some additional yield. A missed default may cost a large part of the principal. **Professionals therefore willingly accept more false alarms to reduce missed defaults.**

Rating migration

Credit quality is not limited to two outcomes. A rating can **migrate** — AA may fall to A, BBB to BB — and every downgrade affects value, because the required risk premium rises and the bond price falls.

A **rating migration matrix** shows the probability of moving from the current rating to another over the next year. Read it by starting at the current rating and reading across the row. Most ratings stay unchanged, which is why the largest probabilities sit along the main diagonal, and weaker ratings show a greater probability of moving down or defaulting.

An AAA borrower might have about a 90% chance of staying, 8.5% of falling to AA, and close to zero of defaulting. A BBB borrower might have about 80% of staying, 9% of falling to BB, and about 2.8% of defaulting. A machine learning model can estimate these probabilities for a *specific* borrower using current company information, which is more useful than an industry-wide average from a published table.

THE CONFUSION MATRIX • 1,000 BONDS, FOUR OUTCOMES

	Bond was actually safe	Bond actually defaulted
Model said safe	940 — correct clearance	5 — missed defaults
Model said risky	42 — false alarms	13 — correctly caught

Accuracy is 95.3%. But of 18 actual defaults, 5 were missed — and a lazy model predicting no defaults at all would score 98%. The four counts matter; the headline percentage does not.

Exercise. A model flags 60 bonds as risky; 9 default, so 51 warnings were false. It clears 940; 4 of those default. Loosening the threshold catches more of the four but creates more false alarms. Most credit managers loosen — because the two mistakes do not cost the same.

Figure 9.1 The confusion matrix. Read the diagonals separately; the total conceals the danger.

KEY IDEA

There is no threshold that minimises both errors at once. Choosing one is a statement about **which mistake you would rather make** — and in credit portfolios, where a missed default costs principal and a false alarm costs yield, the answer is usually to accept more false alarms.

EXAM POINTER

Draw the matrix as a two-by-two grid and label all four cells. Compute accuracy, then immediately note what it conceals — the missed defaults and the lazy-model comparison. For migration, state that the largest probabilities sit on the diagonal and that weaker ratings carry higher downgrade and default probabilities.

QUICK RECAP • TOPIC 9

- The confusion matrix counts four outcomes; the two error types are not equivalent.
- 953 correct out of 1,000 is 95.3% accuracy — yet 5 of 18 actual defaults were missed.
- A lazy model predicting no defaults would score 98% and identify nothing.
- A false alarm costs yield; a missed default costs principal, so professionals accept more false alarms.
- Rating migration matrices give the probability of moving between ratings, with the diagonal dominant.

Module summary, by learning objective

Organised the way assessment will test you, rather than in topic sequence.

LO	What you should now be able to do	Where it was covered
1	Explain the scarcity of bond data, choose a data shape, and clean a raw file	Topics 1, 2 and 3 • Figures 1.1, 2.1, 3.1
2	Estimate a silent bond's yield and state its uncertainty	Topic 4 • Figure 4.1
3	Describe the three yield curve movements and maximum likelihood	Topics 5 and 6 • Figures 5.1, 6.1
4	Explain default classification and read a confusion matrix	Topics 7, 8 and 9 • Figures 7.1, 8.1, 9.1

The module in six sentences

The bond market's problem is the opposite of the equity market's: 240 of Arvind's 268 bonds produced no price at all, yet a value must be published for every one tonight. Finding and cleaning what data does exist takes four-fifths of a real project, and cleaning is part of the modelling rather than preliminary to it — two analysts with the same file and the same algorithm can produce different models through cleaning decisions alone. A bond that did not trade can still be priced by fitting a yield curve through the bonds that did and reading each silent bond's yield off it, provided every estimate is flagged and carries a confidence band. Principal component analysis reduces fifteen moving yields to three drivers — level, slope and curvature — explaining about 97% of movement between them, though the machine finds the components and humans name them. Maximum likelihood chooses the settings that make the observed data most probable, and the sharpness of that likelihood is what produces the confidence bands. And a credit rating is a photograph taken on one day, which a daily model supplements rather than replaces — outputting a probability, never a verdict, and judged by four counts rather than a headline accuracy that a lazy model could beat.

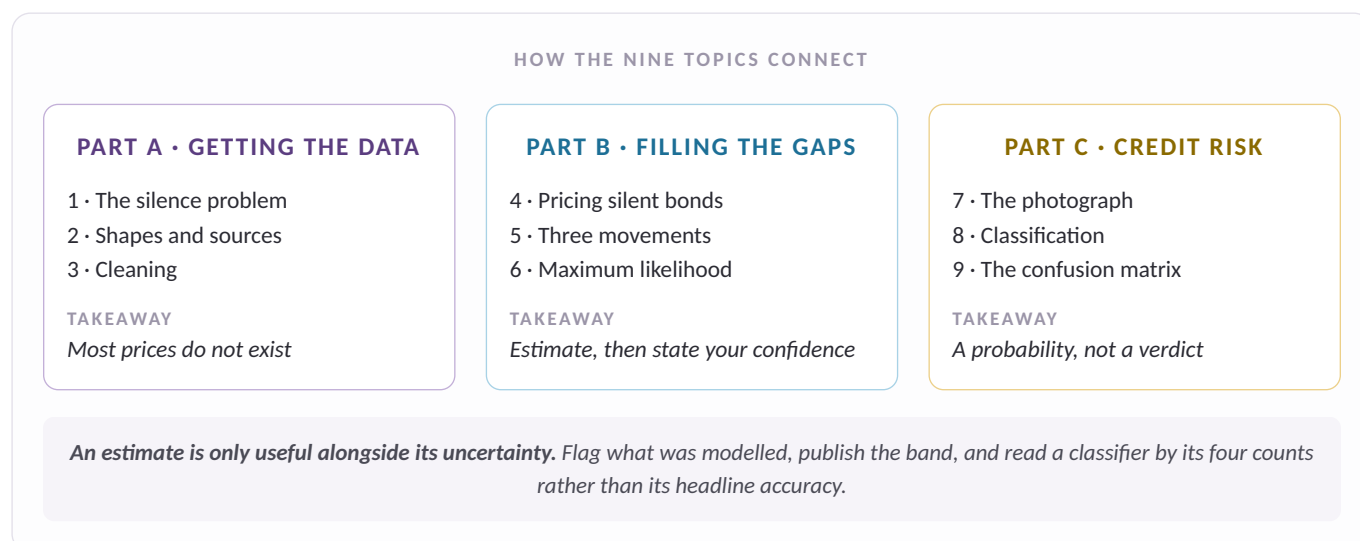


Figure M1 How the nine topics connect. Revise along the parts, not down the list.

Glossary and self-check



Term	Definition
Cross-sectional data	Many items observed at one point in time — a snapshot of the market.
Time series data	One item observed across many dates — the life story of one bond.
Panel data	Many items across many dates; the richest and most demanding structure.
ISIN	International Securities Identification Number — the unique code assigned to a bond.
Web scraping	Programmatically reading a web page to extract data; must respect the site's terms of use.
Covenant	A promise made by the borrower to the lender, whose breach may signal approaching distress.
Matrix pricing	Estimating a silent bond's yield from comparable bonds that did trade.
Imputation	Estimating a missing value using patterns learned from available values.
Yield curve	The yield the market demands at each maturity, from a few months to thirty years.
Bootstrapping	Building the yield curve step by step from the shortest maturity outwards.
Nelson-Siegel	A yield curve formula using few parameters but flexible enough for real shapes.
Confidence band	A range around an estimate showing how certain the model is.
Principal component analysis	An unsupervised technique reducing many related variables to a few drivers.
Level, slope, curvature	The three movements explaining roughly 70%, 20% and 7% of yield curve variation.
Maximum likelihood	Choosing the model settings that make the observed data most probable.
Classification	Supervised learning where the predicted answer is a category rather than a number.
Logistic regression	A classification method using an S-shaped curve to keep probabilities between 0 and 1.
Interest coverage ratio	Profit divided by interest payable; a low value signals difficulty meeting obligations.
Threshold	The predicted probability above which a bond is treated as too risky.
Confusion matrix	A table counting correct classifications, false alarms and missed defaults.
False alarm	A safe bond classified as risky — costs forgone yield.
Missed default	A risky bond classified as safe — costs principal.
Rating migration matrix	The probabilities of moving from one rating to another over the next year.

Self-check

Ungraded retrieval practice. Answer from memory first, then check.

Q1 Why can Arvind not simply carry yesterday's prices forward for the 240 bonds that did not trade?

Because the value being published is tonight's, not last month's. Investors will buy and sell fund units at that value tomorrow morning. Rates may have moved substantially since the last trade, and a stale price misstates the fund's worth without flagging that it does.

Q2 You need to know which warning signs preceded corporate defaults over the last decade. Which data shape do you need, and why will the others not do?

Panel data. You need many companies observed across many dates, so you can trace each borrower's trajectory. A cross-section gives one moment with no history; a time series gives one company with no comparison group.

Q3 A raw file shows twelve bonds with a price of zero. What must you not do, and what are the honest alternatives?

Do not treat zero as a price. A missing price is not zero — doing so collapses the reported portfolio value and teaches the model unreal relationships. The three honest options are: remove the row and disclose it, carry the last observed value forward and label it, or estimate it with a model and flag it as an estimate.

Q4 Three comparable AA five-year bonds traded between 7.85% and 7.95%. A fourth did not trade. What do you report?

About 7.90%, flagged as an estimate, with a narrow confidence band such as 7.88% to 7.92%. This is matrix pricing, or imputation in machine learning language. Because comparable bonds traded recently, the evidence is strong and the band can be narrow — unlike a fifteen-year bond from a small issuer.

Q5 The RBI cuts the repo rate. Short yields fall half a point; thirty-year yields barely move. Which principal component moved, and why might a duration-hedged portfolio still lose?

Slope — the curve steepened. Duration protects mainly against a *level* movement, where the whole curve shifts together. A portfolio long at one end and short at the other can look hedged on duration and still be exposed to a change in slope.

Q6 A default model is 95.3% accurate. Why is that not the number to report?

Because defaults are rare, so accuracy is easy to achieve and conceals what matters. Of 18 actual defaults the model missed 5. A lazy model predicting no defaults at all would score 98% and catch nothing. Report the four counts — correct clearances, correct catches, false alarms and missed defaults — because the two errors carry very different costs.